

Lab Session

Based on Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Ques1:

A factory has two machines. Machine A produces **70%** of the items and has a **4% defect rate**. Machine B produces **30%** of the items and has a **5% defect rate**. If a randomly selected item is found to be defective, what is the probability that it was produced by **Machine A**? (0.6512)

Ques 2:

A disease affects 2% of the population. A medical test detects disease correctly by 95% and false positive rate is 10%. Find the probability that a person actually has disease if test result is positive. (0.162)

Ques 3:

A company has **three suppliers**. Supplier A supplies **50%** of the parts, with **2% defective**. Supplier B supplies **30%** of the parts, with **3% defective**. Supplier C supplies **20%** of the parts, with **5% defective**. From the total production, a part is selected at random and is found to be **defective**. What is the probability that the defective part came from **Supplier C**? (0.345)

Ques4: Load the iris dataset and find

- What is the probability that a flower is Setosa given that petal length is greater than 1.4cm (0.89)
- A flower is chosen with Sepal length is greater than 6cm . What is the probability that chosen flower is Versicolor. (0.27)